

Neval Babu

Right to Work: UK (No sponsorship required) | Leicester | 07459875933 | Neval271@gmail.com

Portfolio: neval-babu-portfolio.netlify.app

EDUCATION

BSc (Hons) Computer Science | University of Leicester | **First Class Honours (76+ %)**

PROJECTS

PitchBrief - Personalised Football News Aggregator | www.pitchbrief.co.uk

- Developed a full-stack personalised football news aggregator using React, TypeScript, Node.js, Express and PostgreSQL, bringing club-specific articles from external RSS feeds into one responsive platform.
- Built Express backend functionality for authentication, favourite clubs and article bookmarks, using JWT-protected routes and bcrypt password hashing to secure personalised features.
- Designed relational PostgreSQL data models with Prisma ORM to provide structured storage for users, clubs, articles and bookmarks.
- Performed integration testing in Postman to verify backend requests, responses, authentication and error handling.
- Configured Vitest to unit-test core functionality, helping identify regressions and confirm that individual features behaved as expected.
- Implemented a dual-deployment architecture with the React frontend on Netlify and the Express backend with PostgreSQL on Railway.

IBM SkillsBuild Companion App

- Developed a gamified web application for IBM SkillsBuild within an 8-member team to create a more engaging learning experience
- Built backend services using Java and Spring Boot, implementing REST APIs to handle business logic and application functionality
- Configured a MySQL relational database for secure data storage and structured user management
- Led the Agile development process as Scrum Master, coordinating sprint activities and team progress to ensure project milestones were met
- Applied Spring Security to support secure authentication and controlled access to application features
- Utilised GitLab for version control and collaboration, supporting efficient teamwork and maintaining code consistency

Portfolio Website | neval-babu-portfolio.netlify.app

- Built and deployed a responsive portfolio using React, TypeScript, Vite and CSS, rendering typed project data through reusable components to keep content consistent and easier to maintain across devices.
- Implemented scroll-aware navigation with IntersectionObserver, persistent motion controls with localStorage, and accessible image galleries using React state, keyboard handling and preloading to prevent layout shifts.
- Configured Playwright end-to-end tests across Chromium, Firefox, WebKit and mobile viewports, validating navigation, motion controls, galleries and responsive layouts to prevent regressions.

FlavorQuest

- Built a responsive Leicester restaurant-discovery website using semantic HTML, CSS and JavaScript, delivering an accessible and consistent experience across desktop and mobile devices.
- Used JavaScript, the Fetch API and DOM manipulation to render restaurant listings from a central JSON dataset and apply combined search, flavour and dietary filters, giving users instant results without page reloads.
- Implemented Progressive Web App functionality with a web app manifest, service worker and Cache API, enabling installation, offline access and faster repeat visits.

Scrabble Game | nb-scrabble.netlify.app

- Built a modular Scrabble game with Python, Pygame and object-oriented programming, supporting local player-versus-player and three computer-opponent difficulty levels for varied gameplay.
- Leveraged Python sets, dictionaries and targeted search algorithms to build an efficient computer-opponent move generator, reducing unnecessary candidate checks.
- Adapted the game for browser deployment with Pygbag/WebAssembly and asynchronous game loops, enabling users to play directly through a web browser.
- Used pytest to create automated unit tests, helping prevent regressions and verify core gameplay logic.

Private Cloud Whiteboard

- Distributed a shared, interactive whiteboard application across multiple Microsoft Azure App Service nodes
- Achieved persistent, low-latency connections by integrating Azure Web PubSub for WebSocket communication and Redis Cache for state synchronisation and rapid data retrieval
- Applied Docker containers to simplify horizontal scaling, allowing quick instance deployment during peak demand

WORK EXPERIENCE

Amazon Warehouse Associate | June 2021 - Present

- Processed and organised shipments with accurate sorting, packing, and labelling to ensure efficient order fulfilment and workflow operations
- Applied strong attention to detail and adaptability to maintain quality in a fast-paced working environment
- Collaborated effectively with team members to support daily productivity targets and operational efficiency

SKILLS

Languages & Frontend: Python, Java, C++, JavaScript, TypeScript, HTML, CSS, React, React Hooks, Vite, Pygame

Backend & Data: Node.js, Express, Spring Boot, Spring MVC, Spring Security, PostgreSQL, MySQL, Redis, Prisma ORM, SQL, REST APIs, WebSockets, RSS, JWT, bcrypt

Testing, Cloud & Tools: Vitest, Integration Testing, Postman, Swagger/OpenAPI, Git, GitHub, GitLab, Docker, Microsoft Azure, Azure App Service, Azure Web PubSub, Netlify, Railway, Android Studio, Shell Scripting

Practices: Agile, Scrum, SDLC, Object-Oriented Programming, Responsive Web Design